AQ4.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

Find the rank of the following matrix: $\begin{bmatrix} 0131 \\ 1001 \\ 3203 \end{bmatrix}$ $\begin{bmatrix} & & & \\ & & & \\ 013 & 102 & 300 & 113 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Consider the following set of vectors SS in R^3R3 to answer the questions 2, 3 and 4.
S = S = $\{(1,2,0),(0,3,1),(3,3,-1),(3,0,-2)(1,2,0),(0,3,1),(3,3,-1),(3,0,-2)\}$.
1 point
Let VV be the vector space spanned by the vectors $(1,2,0)$ and $(0,3,1)(1,2,0)$ and $(0,3,1)$. Which one of the following is correct?

- $V = \{(a, 2a+3b, b) \mid a, b \in R\}$V $= \{(a, 2a+3b, b) \mid a, b \in R\}$
- $V = \{(0, 2a-3b, b) \mid a, b \in R\}$V $= \{(0, 2a-3b, b) \mid a, b \in R\}$
- $V = \{(-a, 2a+3b, 0) \mid a, b \in R\}$V $= \{(-a, 2a+3b, 0) \mid a, b \in R\}$
- $V = \{(0, 2a+3b, 0) \mid a, b \in R\}$V $= \{(0, 2a+3b, 0) \mid a, b \in R\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$V = \{(a, 2a+3b, b) \mid a, b \in R\}$V $= \{(a, 2a+3b, b) \mid a, b \in R\}$
If the vectors in SS are written as the columns of a matrix AA, then what will be the rank of AA?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
What will the dimension of the vector space spanned by SS be?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

1 point
Choose the set of correct options from the following.
- [ ] Row rank and column rank of a matrix is always the same.
- [ ] The rank of a zero matrix is always 0.
- [ ] The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.
- [ ]
Rank of the $n \times n$ identity matrix is 1 for any $n \in N$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Row rank and column rank of a matrix is always the same.
The rank of a zero matrix is always 0.
The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.

Level 2

1 point
Consider the following upper triangular matrix to choose the correct options.

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

where $a, b, c, d, e, f \in R$.
- [ ]
If $f = 0$, then the rank of the matrix must be less than or equal to 2.
- [ ]
If $f = 0$, then the rank of the matrix must be exactly 2.
- [ ]
If a, b, c, d, e, f are all non-zero then the rank of the matrix must be 3.
- [ ]
If a, d, f are non-zero then the rank of the matrix must be 3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $f = 0$, then the rank of the matrix must be less than or equal to 2.
If a, b, c, d, e, f are all non-zero then the rank of the matrix must be 3.
If a, d, f are non-zero then the rank of the matrix must be 3.

If rank of the matrix $\begin{bmatrix} 2 & -3 & 4 \\ 0 & 1 & -2 \\ 1 & -3 & a \end{bmatrix}$ is 2 then find the value of a.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5

1 point

Find the rank of the matrix AA, where $A = [a_{ij}]$A = [aij] is of order $3 \times 3$3 × 3 and $a_{i,j} = \min\{i, j\}$ ai,j = min{i, j}, $i, j = 1, 2, 3$i, j = 1, 2, 3.

[Hint: Write down the matrix and apply row operations to derive it's reduced row echelon form.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

What is the rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{bmatrix}$A = [1xx21yy21zz2], where x, y, zx, y, z are distinct non-zero integers?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Rank of a $3 \times 3$3 × 3 matrix AA is 3 and another $3 \times 3$3 × 3 matrix BB is 2. What is the minimum possible value for the rank of ABAB?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Check Answers

Your score is: 0/10